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URL of this page: https://medlineplus.gov/lab-tests/complete-blood-count-cbc/

Complete Blood Count (CBC)

What is a Complete Blood Count?

A complete blood count, or CBC, is a group of blood tests that measure the number and size of the different cells in your blood. A CBC measures:

- Red blood cells [https://medlineplus.gov/lab-tests/red-blood-cell-rbc-count/] , which carry oxygen from your lungs to the rest of your body.
- White blood cells [https://medlineplus.gov/lab-tests/white-blood-count-wbc/] , which fight infections and other diseases. There are five major types of white blood cells. A CBC test measures the total number of white cells in your blood. A different test called a CBC with differential [https://medlineplus.gov/lab-tests/blood-differential/] measures the number of each type of these white blood cells.
- Platelets [https://medlineplus.gov/lab-tests/platelet-tests/] , which stop bleeding by helping your blood to clot.
- Hemoglobin [https://medlineplus.gov/lab-tests/hemoglobin-test/] , an iron-rich protein in red blood cells that carries oxygen from your lungs to the rest of your body.
- Hematocrit [https://medlineplus.gov/lab-tests/hematocrit-test/] , the amount of whole blood that is made up of red blood cells.
- Mean corpuscular volume (MCV) [https://medlineplus.gov/lab-tests/mcv-mean-corpuscular-volume/] , the average size of your red blood cells.

Other names for a complete blood count: CBC, full blood count, blood cell count

What is it used for?

A complete blood count is often part of a routine checkup [https://medlineplus.gov/healthcheckup.html] . It is also used to monitor a condition or treatment that may affect your blood cell counts such as infections, anemia [https://medlineplus.gov/anemia.html] , immune system disorders [https://medlineplus.gov/immunesystemanddisorders.html] , and blood cancers.

Why do I need a complete blood count?

Your health care provider may have ordered a complete blood count as part of your checkup or to monitor your overall health. The test may also be used to:

- Help diagnose blood diseases [https://medlineplus.gov/blooddisorders.html] , infections, immune system disorders, or other medical conditions
- Check for changes in an existing blood disorder

What happens during a complete blood count?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

Will I need to do anything to prepare for the test?

Usually, no special preparation is necessary for a complete blood count. But if your provider ordered other tests on your blood sample, you may need to fast [https://medlineplus.gov/lab-tests/fasting-for-a-blood-test/] (not eat or drink) for several hours before the test. Your provider will let you know if there are any special instructions.

Are there any risks to the test?

There is very little risk to having a blood test. You may experience slight pain or bruising at the spot where the needle went in, but most symptoms go away quickly.

What do the results mean?

A CBC counts the cells in your blood. There are many reasons your levels **may not be in the normal range**. For example:

- Abnormal levels of red blood cells, hemoglobin, or hematocrit may be a sign of dehydration [https://medlineplus.gov/dehydration.html] , anemia, heart disease [https://medlineplus.gov/heartdiseases.html] , or too little iron [https://medlineplus.gov/iron.html] in your body.
- A low white cell count may be a sign of an autoimmune disorder [https://medlineplus.gov/autoimmunediseases.html] , bone marrow disorder [https://medlineplus.gov/bonemarrowdiseases.html] , or cancer [https://medlineplus.gov/cancer.html] .
- A high white cell count may be a sign of an infection or a reaction to a medicine [https://medlineplus.gov/drugreactions.html] .

If any of your levels are abnormal, it doesn't always mean you have a medical condition that needs treatment. Diet, activity level, medicines, having a menstrual period [https://medlineplus.gov/menstruation.html] , not drinking enough water, and other factors can affect the results. Talk with your provider to learn what your results mean.

Learn more about laboratory tests, reference ranges, and understanding results [https://medlineplus.gov/lab-tests/how-to-understand-your-lab-results/] .

Is there anything else I need to know about a complete blood count?

A complete blood count is only one tool your provider uses to learn about your health. Your provider will consider your medical history, symptoms, and other factors to make a diagnosis. You may also need

additional tests.

References

1. Cleveland Clinic: Health Library: Diagnostics & Testing [Internet]. Cleveland (OH): Cleveland Clinic; c2024. Blood Tests; [reviewed 2022 Dec 06; cited 2024 Jul 17]; [about 16 screens]. Available from: https://my.clevelandclinic.org/health/diagnostics/24508-blood-tests [https://my.clevelandclinic.org/health/diagnostics/24508-blood-tests]
2. Cleveland Clinic: Health Library: Diagnostics & Testing [Internet]. Cleveland (OH): Cleveland Clinic; c2024. Complete Blood Count (CBC); [reviewed 2024 Mar 14; cited 2024 Jul 17]; [about 14 screens]. Available from: https://my.clevelandclinic.org/health/diagnostics/4053-complete-blood-count [https://my.clevelandclinic.org/health/diagnostics/4053-complete-blood-count]
3. El Brihi J, Pathak S. Normal and Abnormal Complete Blood Count With Differential. [Updated 2024 Jun 8]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2024 Jan-[cited 2024 Jul 17]; [about 6 screens]. Available from: https://www.ncbi.nlm.nih.gov/books/NBK604207/ [https://www.ncbi.nlm.nih.gov/books/NBK604207/]
4. Mayo Clinic [Internet]. Mayo Foundation for Medical Education and Research; c1998-2024. Complete Blood Count (CBC); [cited 2024 Jul 17]; [about 6 screens]. Available from: https://www.mayoclinic.org/tests-procedures/complete-blood-count/about/pac-20384919 [https://www.mayoclinic.org/tests-procedures/complete-blood-count/about/pac-20384919]
5. Merck Manual Consumer Version [Internet]. Kenilworth (NJ): Merck & Co., Inc.; c2024. Complete Blood Count (CBC); [cited 2024 Jul 17]; [about 2 screens]. Available from: https://www.merckmanuals.com/home/multimedia/lab-tests/complete-blood-count [https://www.merckmanuals.com/home/multimedia/lab-tests/complete-blood-count]
6. Merck Manual Consumer Version [Internet]. Kenilworth (NJ): Merck & Co., Inc.; c2024. Tables: Complete Blood Count (CBC); [cited 2024 Jul 17]; [about 1 screen]. Available from: https://www.merckmanuals.com/home/multimedia/table/complete-blood-count-cbc [https://www.merckmanuals.com/home/multimedia/table/complete-blood-count-cbc]
7. National Cancer Institute [Internet]. Bethesda (MD): U.S. Department of Health and Human Services; NCI Dictionary of Cancer Terms: complete blood count [cited 2024 Jul 17]; [about 1 screen]. Available from: https://www.cancer.gov/publications/dictionaries/cancer-terms/def/complete-blood-count [https://www.cancer.gov/publications/dictionaries/cancer-terms/def/complete-blood-count]
8. National Cancer Institute [Internet]. Bethesda (MD): U.S. Department of Health and Human Services; NCI Dictionary of Cancer Terms: hematocrit [cited 2024 Jul 17]; [about 1 screen]. Available from: https://www.cancer.gov/publications/dictionaries/cancer-terms/def/hematocrit [https://www.cancer.gov/publications/dictionaries/cancer-terms/def/hematocrit]
9. National Heart, Lung, and Blood Institute [Internet]. Bethesda (MD): U.S. Department of Health and Human Services; Types of Blood Tests; [cited 2024 Jul 17]; [about 6 screens]. Available from: https://www.nhlbi.nih.gov/health-topics/blood-tests [https://www.nhlbi.nih.gov/health-topics/blood-tests]
10. National Heart, Lung, and Blood Institute [Internet]. Bethesda (MD): U.S. Department of Health and Human Services; Your Guide to Anemia; 2021 Sep [cited 2025 Jul 23]; [about 37 screens]. Available from: https://www.nhlbi.nih.gov/sites/default/files/publications/nhlbi-ospeec-your-guide-to-anemia-booklet-release-508.pdf [https://www.nhlbi.nih.gov/sites/default/files/publications/nhlbi-ospeec-your-guide-to-anemia-booklet-release-508.pdf]
11. Nemours KidsHealth [Internet]. Jacksonville (FL): The Nemours Foundation; c1995-2024. Getting a Blood Test; [reviewed 2021 Sep; cited 2024 Jul 17]; [about 4 screens]. Available from: https://kidshealth.org/en/parents/blood-tests.html [https://kidshealth.org/en/parents/blood-tests.html]
12. Testing.com [Internet]. Seattle (WA).: OneCare Media; c2024. CBC Blood Test (Complete Blood Count); [modified 2022 Sep 28; cited 2024 Jul 17]; [about 12 screens]. Available from: https://www.testing.com/tests/complete-blood-count-cbc/ [https://www.testing.com/tests/complete-blood-count-cbc/]

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